MIT ZARAGOZA
International Logistics Program

MZ505 Inventory Systems Fall 2026

Assignment #4

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- **Individual Assignment**
- **Due: Nov 4th (Wed): 08.30 AM**
- **Please submit you work online via Canvas (only one .pdf file) and provide a hard copy before class.**

Problem 1 (Max Points 60)

Weekly demand for 3Ps organic jam at the grocery store of a large retailer in Zaragoza is normally distributed, with a mean of 200 and a standard deviation of 120. The inventory manager continuously monitors inventory and currently orders 1,000 jars of jam each time the inventory position drops to 600. The manufacturer of the organic jam currently takes two weeks to fill an order. (**NOTE**: in this problem use the heuristic approach to continuous review inventory management; i.e., set Q eaqual to EOQ and R based on desired CSL)

(a) How much safety inventory does the retail store carry? (5 points)

(b) What in-stock probability (cycle service level, CSL) does the retail store achieve as a result of this policy? (5 points)

(c) What fill rate (FR) does the store achieve? (5 points)

(d) Calculate CSL and FR again for reorder point (R) of 500, 400, 300, and 200 respectively. What do you observe regarding the impact of R on CSL and FR? (5 points)

(e) It is known that FR is larger than CSL in most cases (but not always) as we observe in (d). Could you explain the reason of it using the definitions of CSL and FR? (5 points)

(f) In contrast to (e), could you come up with a simple counter example where CSL is larger than FR? (Hint: You don't need to assume any policy at all; just develop a case of demand realizations consisting of only a few cycles as in the lecture example) (5 points)

(g~h) The manager now wants a higher target CSL of 95%. (5 points)

(g) What should its new R be? (5 points)

(h) How much safety inventory should the store carry? (5 points)

(i) Now assume that the supply lead time from the manufacturer is normally distributed, with a mean of 2 weeks and standard deviation 1.1 week. How much safety inventory should

the retail store carry if it wants to provide a CSL of 95%, and what's the impact of the variability in lead time on the safety stock? (5 points)

(j~l) Back to (a), the manager now wants to follow a periodic review policy while achieving the same CSL of 95%. The order can be placed every week. (5 points)

(j) What should its OUT level be? (5 points)

(k) How much safety inventory should the store carry? (assume E[back]=0) (5 points)

(l) Compare the safety inventory level of (k) with that of (h) above. Could you explain why they are different although the target CSL is identical? More specifically, why is one higher than the other? (5 points)

Problem 2 (Max Points 30)

Cyber Chemicals uses liquid nitrogen on a regular basis. Average daily demand is 178 gallons with a standard deviation of 45. Due to a substantial ordering cost, which is estimated to be $58 per order (no matter the quantity in the order), Cyber currently orders from its supplier on a weekly basis. Cyber also incurs holding costs on its inventory which is estimated equal to $0.80 per gallon per week. Cyber's supplier delivers in less than a day (i.e., assume that lead time is zero). Assume 52 weeks per year, five days per week. (**NOTE**: this is a periodic review inventory problem)

(a) Cyber wishes to maintain a 99.9% in-stock probability. If it does so, how much safety stock should Cyber carry (use an approximate calculation for SS; i.e., ignore backorders)? (10 points)

(b) What is Cyber's annual ordering cost? (Assume that the probability of having zero demand in a single week is zero). What is its annual inventory holding cost? (10 points)

(c) Should Cyber consider ordering every two weeks? (10 points)

Problem 3 (Max Points 10)

A company sells air filters at a rate of 2,400 per year. The inventory is reviewed continuously and it costs the company $25 to place an order with the distributor of the filters. The holding and backorder (lost sales) costs are estimated to $2 per unit per year and $5 per unit short, respectively. The lead time for filling an order is 2 weeks and the demand during lead time is uniform in [0, 200]. Following the solution procedure for the approximate formulation shown in class, what are the optimal order quantity (Q) and reorder point (R) for the backorder and lost sales cases? What do you observe? What is your intuitive explanation?